PERIPHERALS AND POWER SUPPLY

Day 2: Power Supply Systems

Name: Meghrajsinh Jadeja **Date:** _____

Topic: SMPS, UPS, Power Connectors, and Gaming PC Power Connections

Introduction

An SMPS (Switched Mode Power Supply) changes electricity from a wall socket into the correct DC power needed by computer parts. A UPS (Uninterruptible Power Supply) gives temporary backup power when the main electricity supply fails.

Task 1: Devices That Use SMPS and UPS

A. Three devices that use SMPS

Device	Why it uses SMPS
Desktop computer	It converts AC electricity into DC power for the motherboard, CPU, storage, and other parts.
Wi-Fi router	It needs low-voltage DC power to run its network circuits.
LED monitor	It converts wall electricity into the DC voltages used by the display.

B. Two devices that use UPS

Device	Why it uses UPS
Desktop computer in a computer lab	A UPS gives time to save work and shut down safely during a power cut.
College server or network equipment	A UPS helps keep important services running briefly and prevents sudden shutdowns.

Note: The computer or server is connected to a UPS. The UPS provides backup power to the connected device.

Task 2: Main Internal Components of a Desktop PC SMPS

The diagram below shows the basic flow of power inside a typical PC SMPS. The actual layout may vary by model.

AC INPUT **230 V AC**	→	**RECTIFIER** **AC to DC**	→	**SWITCHING CIRCUIT** **High frequency**
		↓		
DC OUTPUT **12 V / 5 V / 3.3 V**	←	**OUTPUT STAGE** **Filter / regulate**	←	**TRANSFORMER** **Changes voltage**

Power flow: AC input → rectifier → switching circuit → transformer → output stage → regulated DC output.

Component functions

Component	Function
AC input	Receives electricity from the wall socket through the power cable.
Rectifier	Changes incoming AC into DC.
Switching circuit	Switches the DC at high frequency so it can be converted efficiently.
Transformer	Changes the voltage to suitable levels and provides electrical isolation.
Output stage	Filters and regulates the power to make stable DC outputs.
DC output	Sends the required DC power to PC components through the connectors.

Task 3: Power Connectors

Safety rule: Switch off and unplug the PC before connecting or disconnecting cables. Hold the plastic connector, not the wires. Never force a connector into a socket.

Connector name	Devices it powers	One safety tip
ATX 24-pin motherboard connector	Supplies main power to the motherboard.	Align the clip with the socket and press gently until it locks.
SATA power connector	Powers SATA hard drives, SATA SSDs, and some accessories.	Match the L-shaped key before inserting it; never force it.
Molex 4-pin connector	Powers some older drives, fans, and accessories.	Hold the plastic plug when removing it; do not pull the wires.
PCIe 6/8-pin connector	Provides extra power to a graphics card (GPU).	Use the correct PCIe cable, not the CPU/EPS cable.

Connector reference: These are the common connector names to identify during practical work. If your teacher requires photos, add clear photos of the actual connectors beside this table.

Task 4: Connecting the SMPS in a New Gaming PC

1. Prepare safely

Shut down and unplug the PC. Keep the SMPS switch off. Touch an unpainted metal part of the case to reduce static electricity.

2. Mount the SMPS

Place the SMPS in the case compartment and secure it with screws. Make sure its fan has enough ventilation.

3. Connect the motherboard

Connect the 24-pin ATX cable from the SMPS to the large 24-pin socket on the motherboard.

4. Connect CPU power

Connect the 4+4-pin CPU/EPS cable to the CPU power socket near the top of the motherboard. Do not use a PCIe cable here.

5. Connect the graphics card

If the graphics card needs extra power, connect the correct PCIe 6-pin, 8-pin, or manufacturer-specified power cable.

6. Connect storage

Connect a SATA power cable from the SMPS to a SATA HDD or SATA SSD. Connect a separate SATA data cable from the drive to the motherboard.

7. Check and close

Check that each plug is fully seated and cables are away from fans. Close the case before reconnecting electricity.

8. Test the PC

Connect the wall power cable, turn on the SMPS, and start the PC. If anything smells burnt or sounds unusual, switch it off immediately.

Conclusion

SMPS provides the correct power to computer parts, while a UPS provides temporary backup during a power cut. Using the correct connector and following safety steps helps protect the PC components.

Submitted by: Meghrajsinh Jadeja